Chem 220L – Gravimetric Determination of Chloride

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Abstract

The mass percent of chloride in unknown soluble salt #361 was determined by precipitation gravimetry. Three portions of the dried unknown (about 0.5 g each) were dissolved in acidified water and treated with silver nitrate. The AgCl precipitate was digested, kept in the dark, collected in pre-weighed sintered-glass crucibles, washed with dilute nitric acid, and dried to constant mass. After buoyancy corrections and conversion with the gravimetric factor $M_{\mathrm{Cl}}/M_{\mathrm{AgCl}} = 0.24735$, the three results were 54.98 %, 55.02 %, and 55.08 % chloride. The mean was 55.03 % with a standard deviation of 0.050 % (0.091 % RSD), giving a 95% confidence interval of 55.03 % $\pm$ 0.13 % for $n = 3$. An uncertainty analysis shows that weighing uncertainty alone ($\approx \pm 0.013$ to 0.023 percentage points) is smaller than the observed scatter, and that known systematic effects (solubility loss, buoyancy) are each at most a few hundredths of a percentage point. Because the unknown's true composition is not known, the precision is well demonstrated but the accuracy cannot be independently confirmed.

1 Introduction

Gravimetric analysis converts an analyte into a pure, stable compound of known composition that can be isolated and weighed. Because the final measurement is a mass, which can be determined very precisely on an analytical balance, gravimetric methods can be both accurate and precise when the precipitate is pure, completely recovered, and stoichiometric.

In this experiment chloride was determined by precipitating it as silver chloride:

$$\mathrm{Ag^+(aq) + Cl^-(aq) \longrightarrow AgCl(s).} \tag{1}$$

Each mole of AgCl contains one mole of chloride, so the mass of chloride in the sample follows directly from the mass of precipitate. Three replicate determinations of unknown #361 were made so that precision could be evaluated using the mean, standard deviation, and percent relative standard deviation (%RSD), and the results were corrected for air buoyancy.

2 Purpose and Objectives

The purpose was to determine the mass percent of chloride in an impure soluble salt by precipitation as AgCl. The specific objectives were to:

1. quantitatively precipitate chloride as AgCl and collect it in a pre-weighed sintered-glass crucible;
2. wash and dry the precipitate to constant mass;
3. correct the sample and precipitate masses for buoyancy;
4. use the 1:1 mole relationship between AgCl and Cl^- to calculate the mass percent chloride; and
5. evaluate precision and the main sources of uncertainty and error using the mean, standard deviation, %RSD, a confidence interval, and an uncertainty budget.

3 Chemical Background

3.1 Precipitation Equilibrium

Silver chloride is only slightly soluble:

$$\mathrm{AgCl(s)} \rightleftharpoons \mathrm{Ag^+(aq) + Cl^-(aq)}, \qquad K_{\mathrm{sp}} = [\mathrm{Ag^+}][\mathrm{Cl^-}] \approx 1.8 \times 10^{-10}. \tag{2}$$

When the ion product exceeds K_{sp}, AgCl forms. A slight excess of $AgNO_3$ is added to hold the equilibrium chloride concentration extremely low (the common-ion effect) so that precipitation is effectively complete. The stoichiometric relationship used throughout is

$$n_{\mathrm{AgCl}} = n_{\mathrm{Cl^-}}. \tag{3}$$

3.2 Colloids, Coagulation, Digestion, and Peptization

Freshly formed AgCl is a fine colloidal suspension whose particles can pass through the pores of a sintered-glass crucible. Charged particles repel one another and stay dispersed. Three steps counter this. First, HNO_3 and the small excess of $AgNO_3$ provide an electrolyte that reduces electrostatic repulsion so the particles can *coagulate*. Second, the mixture is heated to steaming (without boiling or stirring) for 10–20 minutes; this *digestion* lets small particles redissolve and redeposit on larger ones, giving a more filterable solid. Third, the wash solution contains dilute HNO_3, because washing with pure water removes the electrolyte and can cause *peptization*, in which a coagulated precipitate redisperses into colloidal particles that pass through the filter. Peptization loses AgCl, so the calculated chloride would be too low.

3.3 Washing and Constant Mass

The precipitate was washed to remove excess $AgNO_3$, nitrate, acid, and other soluble material. The final filtrate was tested with 1 M HCl; cloudiness would indicate remaining Ag^+, so washing continued until the test stayed clear. The crucible and precipitate were dried at 110 °C, cooled in a desiccator, and reweighed until consecutive masses agreed within 0.4 mg, which ensures that residual water or wash solution is not weighed as product.

3.4 Buoyancy Correction

A balance compares an object with reference weights while both experience air buoyancy. The object's true mass is

$$m_{\mathrm{corr}} = m_{\mathrm{obs}} \left(\frac{1 - \rho_{\mathrm{air}}/\rho_{\mathrm{std}}}{1 - \rho_{\mathrm{air}}/\rho_{\mathrm{obj}}} \right), \tag{4}$$

with $\rho_{\mathrm{air}} = 0.001\,\mathrm{g\,mL^{-1}}$, $\rho_{\mathrm{std}} = 8.0\,\mathrm{g\,mL^{-1}}$, $\rho_{\mathrm{AgCl}} = 5.56\,\mathrm{g\,mL^{-1}}$, and $\rho_{\mathrm{sample}} = 2.17\,\mathrm{g\,mL^{-1}}$ (the density of NaCl, used for the unknown).

3.5 Conversion to Chloride

With $M_{\mathrm{AgCl}} = 143.32\,\mathrm{g\,mol^{-1}}$ and $M_{\mathrm{Cl}} = 35.45\,\mathrm{g\,mol^{-1}}$ the gravimetric factor is $35.45/143.32 = 0.24735$, so

$$m_{\mathrm{Cl^-}} = m_{\mathrm{AgCl,\,corr}} \left(\frac{M_{\mathrm{Cl}}}{M_{\mathrm{AgCl}}} \right), \qquad \%\mathrm{Cl^-} = \frac{m_{\mathrm{Cl^-}}}{m_{\mathrm{sample,corr}}} \times 100. \tag{5}$$

4 Hypothesis

If chloride in unknown #361 is quantitatively precipitated as AgCl, and the precipitate is completely transferred, washed, dried to constant mass, and weighed, then the three replicate mass-percent chloride values will agree closely, giving a small standard deviation and a low %RSD.

5 Materials and Equipment

Chemicals: unknown soluble salt #361; approximately 0.2 M $AgNO_3$; concentrated HNO_3 (15.8 M); 6 M HNO_3; 6 M aqueous NH_3; 1 M HCl; purified water.

Equipment: three sintered-glass filtering crucibles; vacuum flask, adapter, and tubing; analytical balance and weighing bottle; three 400 mL beakers, watch glasses, and stirring rods; hot plate; drying oven (110 °C); desiccator; rubber policeman, spatula, crucible tongs, and tissues.

6 Safety and Waste Disposal

Goggles, a lab coat, and chemical-resistant gloves were worn. Concentrated HNO_3 and HCl are corrosive, and HNO_3 is also a strong oxidizer; they and the NH_3 solution were handled in the fume hood. $AgNO_3$ is corrosive, oxidizing, and stains skin and clothing. Hot plates, the oven, hot crucibles, and heated glassware are burn hazards, so tongs were used and crucibles were cooled before entering the desiccator. The vacuum apparatus was checked before use to reduce the risk of glassware breakage. AgCl is light sensitive:

$$2\,\mathrm{AgCl(s)} \xrightarrow{h\nu} 2\,\mathrm{Ag(s)} + \mathrm{Cl_2(g)}, \tag{6}$$

so the precipitate was covered and stored in the dark. All silver-containing solutions, washings, and solids were collected in the silver waste container; acid and base wastes went to their designated containers, and nothing was poured down the sink unless the instructions permitted it.

7 Experimental Procedure

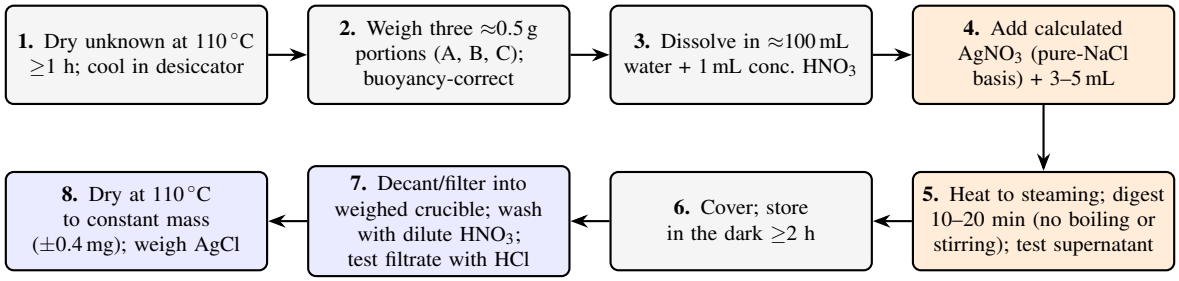


Figure 1: Workflow of the gravimetric chloride determination.

7.1 Preparation of Crucibles

Three sintered-glass crucibles were labeled I, II, and III. About 5 mL of concentrated HNO_3 was added to each and left about 10 minutes in the hood (acid waste collected), after which each crucible was rinsed under vacuum with several portions of tap water. About 5 mL of 6 M NH_3 was added to each and left about 10 minutes (base waste collected), and the crucibles were rinsed with tap water and then three portions of purified water under vacuum. The crucibles were heated at 110 °C for at least 1 h, cooled briefly, then cooled to room temperature in a desiccator with the lid slightly ajar at first, and weighed. Heating, cooling, and weighing were repeated until two consecutive masses differed by no more than 0.4 mg.

7.2 Preparation of the Unknown

About 1.5 g of unknown #361 (recorded: 1.4999 g) was dried in a weighing bottle at 110 °C for at least 1 h and cooled in a desiccator. Three portions of about 0.5 g were weighed to 0.0001 g into separate 400 mL beakers (A, B, C), and the masses were corrected for buoyancy. Each portion was dissolved in about 100 mL of purified water containing 1 mL of 15.8 M HNO_3.

7.3 Precipitation and Digestion

The volume of 0.2 M $AgNO_3$ needed for each portion was calculated assuming the sample was pure NaCl, and about 200 mL of 0.2 M $AgNO_3$ was prepared (0.04 mol, 6.7948 g $AgNO_3$), roughly 50 mL more than the 127.86 mL total required. With good stirring the calculated volume was added slowly to each sample using a separate stirring rod for each beaker, followed by an extra 3–5 mL. Each solution was heated until steaming and digested for 10–20 minutes without stirring or boiling. A few drops of $AgNO_3$ were added to the clear supernatant to test for completeness; any cloudiness would have been treated with about 3 mL more $AgNO_3$ and a repeat digestion. The covered beakers were stored in the dark for at least 2 hours.

7.4 Filtration, Washing, and Drying

About 1 L of wash solution was prepared from purified water and 2–5 mL of 6 M HNO_3. Each supernatant was decanted through its corresponding crucible (sample A with crucible I, B with III, C with II). The precipitate was washed several times in the beaker and transferred quantitatively with small streams of wash

solution; a rubber policeman and small spatula recovered particles from the beaker walls. The final filtrate was tested with 1 M HCl and washing continued until it stayed clear. The crucibles were dried at 110 °C for at least 1 h, cooled in a desiccator, and weighed, repeating until consecutive masses agreed within 0.4 mg.

8 Results

8.1 Crucible Constant-Mass Data

Table 1: Constant-mass determination for the empty crucibles (from the laboratory notebook). The criterion was a change of no more than 0.4 mg between consecutive weighings.

Crucible	Weighing	Mass (g)	Change (mg)	Meets criterion?
I	1	16.4370	–	–
	2	16.4368	0.2	Yes
II	1	20.1263	–	–
	2	20.1257	0.6	No
	3	20.1254	0.3	Yes
III	1	16.4810	–	–
	2	16.4806	0.4	Yes

The second weighing of crucible II changed by 0.6 mg, so it was heated and weighed again; the third weighing agreed within 0.3 mg. The final masses used were 16.4368 g (I), 20.1254 g (II), and 16.4806 g (III).

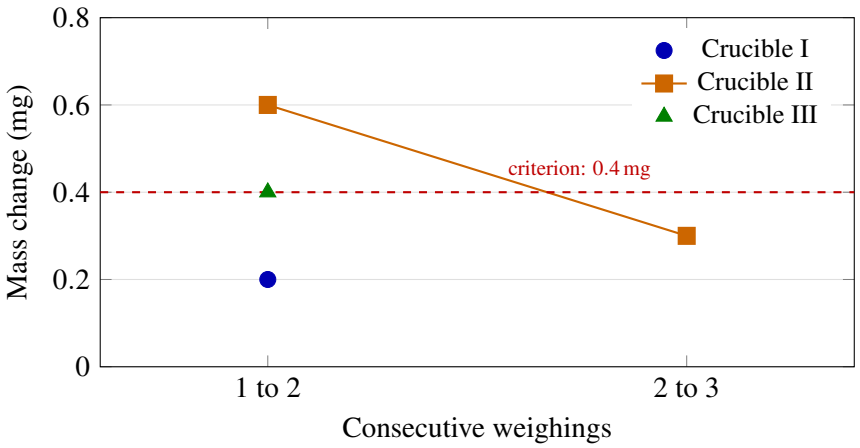


Figure 2: Change in crucible mass between consecutive weighings. Only crucible II needed a third weighing.

8.2 Sample Masses and Silver Nitrate Volumes

Table 2: Sample masses and calculated $AgNO_3$ volumes (pure-NaCl assumption, 0.2 M $AgNO_3$, $M_{NaCl} = 58.44\,g\,mol^{-1}$).

Sample	Crucible	Mass (g)	Corrected (g)	$AgNO_3$ (mL)
A	I	0.4999	0.5001	42.79
B	III	0.5003	0.5005	42.82
C	II	0.4936	0.4938	42.25
			Total	127.86

8.3 Gravimetric Results

Table 3: Masses of AgCl and chloride results for unknown #361.

Sample	Crucible + AgCl (g)	AgCl obs. (g)	AgCl corr. (g)	Cl^- (g)	Cl^- (mmol)	% Cl^-
A	17.5484	1.1116	1.1117	0.2750	7.757	54.98
B	17.5938	1.1132	1.1133	0.2754	7.768	55.02
C	21.2250	1.0996	1.0997	0.2720	7.673	55.08

Table 4: Summary statistics for the three replicates (values rounded to four significant figures at each step, per the laboratory procedure).

Statistic	Value
Mean	55.03 %
Standard deviation (s)	0.050 %
%RSD	0.091 %
Range	0.10 %
95% confidence interval ($t = 4.303$, $n = 3$)	$55.03\,\% \pm 0.13\,\%$

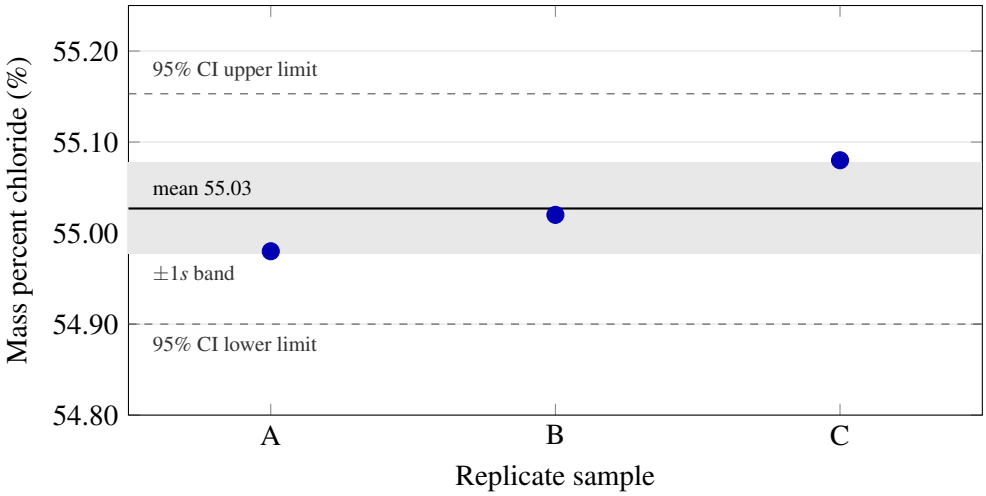


Figure 3: Replicate chloride results. The shaded band is the mean ± 1 standard deviation, and the dashed lines are the 95% confidence limits. The y-axis is truncated (it does not start at zero) to make the small scatter visible.

9 Calculations

Calculations are shown for sample A; B and C were treated identically.

9.1 Buoyancy Correction of the Sample Mass

$$m_{\text{sample,corr}} = 0.4999\,\text{g}\left(\frac{1-0.001/8.0}{1-0.001/2.17}\right) = 0.4999\,\text{g}(1.000336) = \boxed{0.5001\,\text{g}}.$$

9.2 Silver Nitrate Volume (Pure-NaCl Basis)

$$n_{\text{NaCl}} = \frac{0.5001\,\text{g}}{58.44\,\text{g mol}^{-1}} = 8.557\times10^{-3}\,\text{mol}, \qquad V_{\text{AgNO}_3} = \frac{8.557\times10^{-3}\,\text{mol}}{0.2\,\text{mol L}^{-1}} = \boxed{42.79\,\text{mL}}.$$

9.3 Observed and Corrected Mass of Silver Chloride

$$m_{\text{AgCl, obs}} = 17.5484\,\text{g} - 16.4368\,\text{g} = 1.1116\,\text{g},$$
$$m_{\text{AgCl, corr}} = 1.1116\,\text{g}\left(\frac{1-0.001/8.0}{1-0.001/5.56}\right) = \boxed{1.1117\,\text{g}}.$$

9.4 Mass of Chloride and Mass Percent

$$m_{\mathrm{Cl^-}} = 1.1117\,\mathrm{g}\left(\frac{35.45\,\mathrm{g\,mol^{-1}}}{143.32\,\mathrm{g\,mol^{-1}}}\right) = \boxed{0.2750\,\mathrm{g}},$$
$$\%\mathrm{Cl^-} = \frac{0.2750\,\mathrm{g}}{0.5001\,\mathrm{g}} \times 100 = \boxed{54.98\,\%}.$$

9.5 Mean, Standard Deviation, and %RSD

$$\bar{x} = \frac{54.98 + 55.02 + 55.08}{3} = \boxed{55.03\,\%},$$
$$s = \sqrt{\frac{(54.98-55.03)^2 + (55.02-55.03)^2 + (55.08-55.03)^2}{3-1}} = \boxed{0.050\,\%},$$
$$\%\mathrm{RSD} = \frac{0.050}{55.03} \times 100 = \boxed{0.091\,\%}.$$

Carrying unrounded intermediate values instead gives $\bar{x} = 55.03\%$, $s = 0.051\%$, and $\%\mathrm{RSD} = 0.093\%$, the same within rounding.

9.6 Confidence Interval

$$\mathrm{CI}_{95} = \bar{x} \pm \frac{t\,s}{\sqrt{n}} = 55.03 \pm \frac{(4.303)(0.050)}{\sqrt{3}} = \boxed{55.03 \pm 0.13\ \%}.$$

With only $n = 3$ the t-value is large, so the interval is much wider than the standard deviation. Dixon's Q for the extreme value (sample C) is $Q = (55.083 - 55.017)/(55.083 - 54.983) = 0.65$, below the critical value of 0.970 (95%, $n = 3$), so no replicate is an outlier.

10 Data Analysis

10.1 Silver Nitrate Stoichiometry

Because the pure-NaCl assumption overestimates the chloride in a sample that is only about 55 % chloride, the calculated $AgNO_3$ volumes already exceeded the amount actually required (Table 5), before the extra 3–5 mL was added. The reaction therefore had an excess of Ag^+ of at least about 10 %, well beyond a "slight" excess. This favors complete precipitation but increases the importance of thorough washing, because $AgNO_3$ left in the solid would add mass and bias the result high.

Table 5: $AgNO_3$ required by the measured chloride versus the volume calculated on a pure-NaCl basis (before the extra 3–5 mL).

Sample	Cl^- found (mmol)	Required (mL)	Added, calc. (mL)	Excess (%)
A	7.757	38.78	42.79	10.3
B	7.768	38.84	42.82	10.3
C	7.673	38.36	42.25	10.1

10.2 Implied Composition of the Unknown

If NaCl were the only chloride-containing component, a pure sample would be $35.45/58.44 = 60.66\,\%$ chloride, so the measured value corresponds to about $55.03/60.66 = 90.7\,\%$ NaCl-equivalent by mass. This is only an interpretation: the identity of the cation(s) in the unknown is not known, and Figure 4 shows that the measured value is also consistent with other chlorides or mixtures (for example, a mixture of NaCl and KCl would need about 57 % NaCl by moles of salt to give this chloride content, assuming only those two salts). Gravimetric chloride alone cannot identify the unknown.

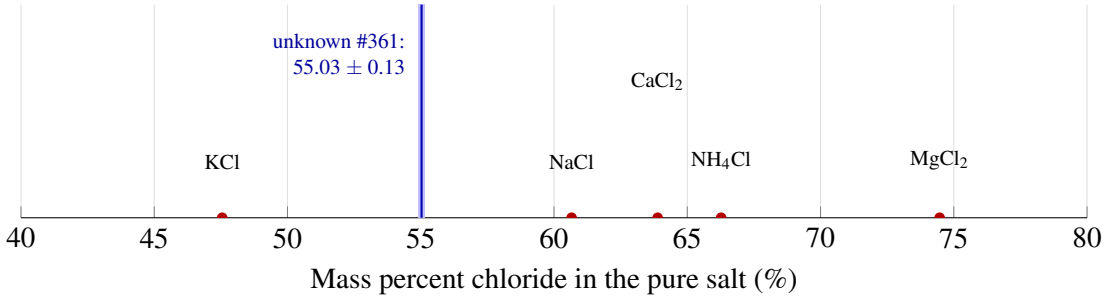


Figure 4: Chloride content of some pure chlorides compared with the measured value for the unknown (blue line with its 95% confidence band). The unknown contains less chloride than pure NaCl, consistent with an impure sample.

10.3 Uncertainty Budget

Weighing uncertainty was estimated from the balance readability (± 0.0001 g per reading, so $\pm 0.000\,14$ g for a mass obtained as a difference of two readings), and, separately, from the 0.4 mg constant-mass tolerance. For sample A,

$$\frac{u(\%\mathrm{Cl}^-)}{\%\mathrm{Cl}^-} = \sqrt{\left(\frac{u_{\mathrm{AgCl}}}{m_{\mathrm{AgCl}}}\right)^2 + \left(\frac{u_{\mathrm{sample}}}{m_{\mathrm{sample}}}\right)^2} = \sqrt{\left(\frac{0.00014}{1.1117}\right)^2 + \left(\frac{0.0001}{0.5001}\right)^2} = 2.4 \times 10^{-4},$$

so $u = 2.4 \times 10^{-4} \times 54.98 = 0.013\,\%$ (percentage points). Using 0.0004 g for the AgCl term gives 0.023 %.

Table 6: Estimated size of individual uncertainty contributions and systematic effects on the mean % Cl^- (percentage points).

Source	Effect	Direction
Balance readability (random)	$\approx \pm 0.013$	either
Constant-mass tolerance, 0.4 mg (bound)	$\approx \pm 0.023$	either
Observed replicate scatter (s)	0.050	either
Solubility of AgCl during washing (upper bound)[a]	≤ 0.03	low
AgCl solubility in the presence of excess Ag^+[b]	$< 10^{-6}$	negligible
Omitting both buoyancy corrections[c]	$+0.02$	high

[a] Assumes about 333 mL of wash per sample fully saturated ($\approx 1.3 \times 10^{-5}$ M, 0.62 mg AgCl).

[b] $[Cl^-] = K_{sp}/[Ag^+] \approx 1.8 \times 10^{-10}/0.011 \approx 1.7 \times 10^{-8}$ M in about 150 mL.

[c] Uncorrected masses give a mean of about 55.05% versus 55.03%.

The observed standard deviation (0.050) is larger than the weighing contributions (0.013–0.023), so the replicate scatter is not explained by the balance alone. The remaining scatter most likely comes from small, uncorrelated differences between samples: precipitate lost or gained during transfer and washing, incomplete drying, and real inhomogeneity of the solid unknown. The systematic effects that can be bounded are each at most a few hundredths of a percentage point, comparable to the random scatter, so they do not change the conclusions.

10.4 Sources of Error

Table 7: Sources of error, their effect on the AgCl mass, and how each was controlled.

Source	Effect	Control	Result
Insufficient electrolyte (peptization)	AgCl redisperses into colloidal particles that pass through the filter or stay in the filtrate	Wash with acidified HNO_3, avoid excess pure water	Low
Insufficient digestion	Particles stay small and are hard to filter completely	Heat without boiling 10–20 min	Low
Incomplete precipitation	Some Cl^- stays dissolved	Slight excess of $AgNO_3$; test the clear supernatant	Low
Excess $AgNO_3$ or poor washing	Soluble $AgNO_3$ or other ions remain with the solid	Moderate excess; wash until the filtrate shows no AgCl cloudiness with HCl	High
Photodecomposition	AgCl forms Ag(s) and Cl_2 in strong light; with excess Ag^+ the liberated chlorine can reprecipitate as extra AgCl	Keep covered and in the dark	High
Incomplete drying	Water or solution is weighed as product	Dry, cool, and reweigh to ± 0.4 mg	High
Loss during transfer/filtration	AgCl stays on beaker, rod, or policeman, or passes through the filter	Quantitative transfer, policeman, careful decanting	Low

11 Discussion

Chloride in unknown #361 was converted to a weighable form, and the three replicates agreed closely: 54.98 %, 55.02 %, and 55.08 %, a spread of only 0.10 %. The mean of 55.03 %, standard deviation of 0.050 %, and %RSD of 0.091 % indicate excellent precision, supporting the hypothesis. The constant-mass procedure contributed to this: all three crucibles satisfied the 0.4 mg criterion (crucible II after an additional heating cycle), so the empty-crucible masses were not drifting with absorbed moisture.

Precision is not accuracy. Because the unknown was an impure salt of unknown composition, there is no reference value against which to judge the result, and a precise but biased result would look identical in the replicate data. The error analysis suggests that the main systematic biases are small. Solubility losses are at most about 0.03 percentage points even under the worst-case assumption that all wash solution was saturated, and the equilibrium solubility in the presence of excess Ag^+ is negligible. Omitting the buoyancy corrections would change the mean by about 0.02 percentage points, so they matter only at the level of the last reported digit. The most important chemical risks are those in Table 7: peptization, incomplete transfer, and incomplete drying or washing. Washing losses and transfer losses bias the result low, whereas occluded $AgNO_3$, residual moisture, and photodecomposition in the presence of excess Ag^+ bias it high. The roughly 10 % excess $AgNO_3$ that resulted from the pure-NaCl volume estimate makes the high-bias risks from inadequate washing more relevant than usual, which is why the HCl test of the final filtrate was important.

Because only three replicates were measured, the 95% confidence interval (± 0.13 %) is wider than the standard deviation suggests. The observed scatter exceeds what balance readability alone would produce, so

sample-to-sample differences in transfer, washing, and drying, and possibly inhomogeneity of the unknown, contributed. Additional replicates would narrow the interval and make outlier tests meaningful. The final result also shows that the unknown contains less chloride than pure NaCl (60.66 %), as expected for an impure sample, but does not identify it.

12 Conclusion

The chloride content of unknown soluble salt #361 was determined by precipitation as AgCl, vacuum filtration, washing with acidified solution, and drying to constant mass. After buoyancy correction the three replicates were 54.98 %, 55.02 %, and 55.08 %, giving

$$\boxed{\%\mathrm{Cl}^- = 55.03\,\% \pm 0.050\,\%\ (1s), \qquad 55.03\,\% \pm 0.13\,\%\ (95\%\ \mathrm{CI},\ n = 3)}$$

with a %RSD of 0.091 %. The hypothesis was supported: the replicates were closely clustered. The uncertainty analysis showed that weighing uncertainty and the bounded systematic effects were each only a few hundredths of a percentage point, and that the measured chloride content is about 90.7 % of that of pure NaCl. To improve a future determination, the $AgNO_3$ volume should be based on a more realistic chloride estimate to avoid a large excess, successive weighings of the AgCl crucibles should be recorded for every drying cycle, and more replicates (or a known standard analyzed alongside) should be run to assess accuracy.

References

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3. Skoog, D. A.; West, D. M.; Holler, F. J.; Crouch, S. R. *Fundamentals of Analytical Chemistry*, 9th ed.; Cengage: Belmont, CA, 2014.